

I didn't build a
while-loop

— I built a state machine —

**It cannot finish
without being checked.**

look inside →

1 IT PLANS FIRST

Before it touches a file it writes the steps down — and the FSM won't let it act until the plan exists.



ai-agent – anatomy

```
$ ai-agent run "test_zero fails: divide(5,0) should return None. Fix calc.py so the tests pass." \  
-w ~/demo/anatomy
```

```
FSM idle --start--> planning
```

```
Plan created (204 chars)
```

Plan

1. Open calc.py to locate the divide function.
2. Modify divide to return None when b is 0.
3. Verify the change by running the tests.
4. Confirm the fix by checking test_zero output.
5. Finish the task.

```
FSM planning --plan_ready--> executing
```

2 IT CAN'T SKIP A CHECK

Success, retry, give-up: every path runs through evaluating. There is no edge from executing to done.



ai-agent – anatomy

```
# A run that works:
```

```
FSM idle      --start-----> planning
FSM planning  --plan_ready-----> executing
FSM executing --execution_done--> evaluating
FSM evaluating --passed-----> done
```

```
# A run that doesn't:
```

```
FSM executing --execution_done--> evaluating
FSM evaluating --failed-----> reflexing
FSM reflexing  --retry-----> executing
FSM executing  --execution_done--> evaluating
FSM evaluating --failed-----> reflexing
FSM reflexing  --give_up-----> error
```

```
# Every path goes through `evaluating`.
```

```
# There is no edge from EXECUTING to DONE.
```

3 IT EDITS, NOT REWRITES

Two lines added, and all 6 original lines byte-identical. It never re-writes your file from memory.



ai-agent – anatomy

```
step 2 → search_replace {"path": "calc.py",
                          "search": "    return a / b",
                          "replace": "    if b == 0:\n        return None\n    return a / b"}
```

Applied edit to calc.py:

--- a/calc.py

+++ b/calc.py

@@ -3,4 +3,6 @@

```
def divide(a, b):
+   if b == 0:
+       return None
    return a / b
```

```
# 2 lines added. All 6 original lines byte-identical --
# it never re-wrote the file from memory.
```

4 IT NOTICES ITSELF

Same tool, same arguments, three times over. It stops and moves on instead of burning your evening.



ai-agent – anatomy

```
$ ai-agent run "Create solver.py with solve(n) so test_solver.py passes." \
-w ~/demo/stuck --max-retries 1
```

```
step 1 → write_file {"path": "solver.py", ...}
```

```
step 2 → write_file {"path": "solver.py", ...}
```

```
Redundant repeat of write_file (x1) at step 2
```

```
...
```

```
step 5 → write_file {"path": "solver.py", ...}
```

```
Redundant repeat of write_file (x2) at step 5
```

```
step 6 → write_file {"path": "solver.py", ...}
```

```
Redundant repeat of write_file (x3) at step 6
```

```
Repeated actions without progress; moving to evaluation.
```

```
No-progress loop detected at step 6; aborting phase
```

```
FSM executing --execution_done--> evaluating
```

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— you are here —

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github.com/Natarajan-R/local-ai-coding-agent

Natarajan Ramasamy